

Genome wide *in silico* characterization of *Dof* gene families of pigeonpea (*Cajanus cajan* (L) Millsp.)

N. Malviya · S. Gupta · V. K. Singh ·
M. K. Yadav · N. C. Bisht · B. K. Sarangi ·
D. Yadav

Received: 9 February 2014 / Accepted: 14 October 2014 / Published online: 26 October 2014
© Springer Science+Business Media Dordrecht 2014

Abstract The DNA binding with One Finger (Dof) protein is a plant specific transcription factor involved in the regulation of wide range of processes. The analysis of whole genome sequence of pigeonpea has identified 38 putative *Dof* genes (*CcDof*) distributed on 8 chromosomes. A total of 17 out of 38 *CcDof* genes were found to be intronless. A comprehensive *in silico* characterization of *CcDof* gene family including the gene structure, chromosome location, protein motif, phylogeny, gene duplication and functional divergence has been attempted. The phylogenetic analysis resulted in 3 major clusters with closely related members in phylogenetic tree revealed common motif distribution. The *in silico cis*-regulatory element

analysis revealed functional diversity with predominance of light responsive and stress responsive elements indicating the possibility of these *CcDof* genes to be associated with photoperiodic control and biotic and abiotic stress. The duplication pattern showed that tandem duplication is predominant over segmental duplication events. The comparative phylogenetic analysis of these Dof proteins along with 78 soybean, 36 *Arabidopsis* and 30 rice Dof proteins revealed 7 major clusters. Several groups of orthologs and paralogs were identified based on phylogenetic tree constructed. Our study provides useful information for functional characterization of *CcDof* genes.

Electronic supplementary material The online version of this article (doi:10.1007/s11033-014-3797-y) contains supplementary material, which is available to authorized users.

N. Malviya · S. Gupta · D. Yadav (✉)
Department of Biotechnology, D.D.U. Gorakhpur University,
Gorakhpur 273 009, Uttar Pradesh, India
e-mail: dinesh_yad@rediffmail.com

V. K. Singh
Centre for Bioinformatics, School of Biotechnology, Banaras
Hindu University, Varanasi 221 005, Uttar Pradesh, India

M. K. Yadav
Department of Biotechnology, SVP University of Agriculture &
Technology, Meerut 250 110, Uttar Pradesh, India

N. C. Bisht
National Institute of Plant Genome Research JNU Campus,
New Delhi 110 067, India

B. K. Sarangi
Environmental Biotechnology Division, National Environmental
Engineering Research Institute- CSIR, Nehru Marg,
Nagpur 440 020, India

Keywords Dof (DNA binding with One Finger) ·
Transcription factor · Pigeonpea · Duplication ·
Phylogeny · *Cis*-regulatory elements

Introduction

Transcriptional regulation in higher plants is based on processes common to all multicellular organisms. The proteins implicated are generally recognizable on the basis of sequence homologies with orthologs in other eukaryotes. The Dof (DNA-binding with one finger) transcription factor is a plant specific transcription factor involved with diverse functions [1]. The first gene containing a conserved Dof domain was isolated from maize [2] and later on identified in several species, revealing multifarious roles specific to plants [3–6]. The *Dof* genes have been reported from many angiosperms, gymnosperms, lower plants including the green alga *Chlamydomonas reinhardtii* and the moss *Physcomitrella patens* [7]. The completed or nearly completed genome sequences information of many vascular and non-vascular plants has provided an

opportunity to fish out *Dof* gene families. Using bioinformatics tools, a total of 36, 30, 41, 26, 31, 28, 78, 54, 27 and 34 *Dof* genes were reported in *Arabidopsis*, rice, poplar, barley, wheat, sorghum, soybean, maize, *Brachypodium* and tomato, respectively [7–15].

The *Dof* transcription factors (TF) are characterized by a highly conserved DNA binding *Dof* domain of 52 amino-acid residues including a C2–C2 type zinc finger motif at its N-terminal region binding specifically to DNA sequences with a T/AAAAG core [16]. This motif is considered to be essential for DNA binding and mutation of any of the four cysteine residues abolishes DNA binding in vitro and in vivo [17]. Both the loop length and the presence of a Zn^{2+} ion directly influence the structure and nature of the DNA binding [1]. The *Dof* domain is known to be a bi-functional domain that mediates not only DNA-binding but also protein–protein interactions including interaction with another class of transcription factors like basic domain-leucine zipper protein, bZIP protein MYBR2R3, MYBR1 and WRKY [5, 18–21] and non-histone nuclear HMG protein [6, 22–26].

The *Dof* transcription factors are implicated in a wide range of processes unique to plants. The *Dof* proteins are involved in regulation of endosperm-specific seed storage proteins [5, 27–31], morphogenesis [32, 33], seed germination [34–38], secondary metabolism [39], root storage development [40], cell wall gene expression [41], defense gene [42, 43], cell cycle [43], phytochrome and cytochrome signaling [44], nitrogen use efficiency [45, 46], the adaxial–abaxial polarity [47] and stomatal functioning [48].

Several *Dof* TFs have been functionally characterized in *Arabidopsis*, rice and others but still the function of most of the predicted *Dof* genes needs to be deciphered. The pigeonpea (*Cajanus cajan* (L) Millsp.) is an important grain legume of the Indian subcontinent, South-East Asia and East Africa. More than eighty five percent of the world pigeonpea is produced and consumed in India where it is considered to be a key crop for food and nutritional security of the people. Pigeonpea belongs to subtribe *Cajaninae* of the *Phaseoleae* under sub-family *Papilionoideae* of family *Leguminosae*. The *C. cajan* is the only domesticated species under sub-tribe *Cajaninae* [49]. The deciphering of whole genome sequence of pigeonpea recently facilitates the genome wide characterization of *Dof* gene family [50, 51].

This paper reports characterization of 38 *Dof* genes with emphasis on putative *CcDof* gene structures, motif analysis, chromosomal distribution, conserved motifs, duplication status, ancestral protein sequence and distribution of various *cis*-elements. Further its comparative phylogeny with soybean, rice and *Arabidopsis* *Dof* gene family representing a legume, model monocot and dicot crop has also been attempted. The putative functions of the predicted *CcDof* genes were also investigated by analyzing the *cis*-regulatory elements associated with these genes in the promoter region.

Materials and methods

Database searches for the identification of *Dof* gene family members in pigeonpea

To identify all members of the *Dof* family in pigeonpea (*Cajanus cajan*), the *Dof* sequences of *Arabidopsis thaliana* and *Oryza sativa* were collected from the DATF (<http://datf.cbi.pku.cn>) [52] database and DRTF (<http://drtf.cbi.pku.cn>) [53] database respectively. The amino-acid sequence of the *Dof* domain was used to search for potential *Dof* domain homolog hits in the whole-genome sequence of *C. cajan* with BLASTP and TBLASTN [54] at the NCBI database (<http://www.ncbi.nlm.nih.gov>) [55]. All non-redundant hits with expected values $<1\text{e-}5$ were collected. The 2 kb upstream and 2 kb downstream sequences of *Dof* domain homologs were retrieved from the whole genome shotgun sequence of *C. cajan* for fishing out the putative *CcDof* genes. As for the incorrectly-predicted genes, manual re-annotation was performed using the on-line web server GENSCAN (<http://genes.mit.edu/GENSCAN.html>) [56] and FGENESH (<http://linux1.softberry.com/berry.phtml>) [57]. The putative *Dof* protein sequences of *C. cajan* were subjected to protein functional analysis using PFAM version 27.0 [58], PROSITE 20.93 [59], INTERPROSCAN version 4.0 (www.ebi.ac.uk/Tools/pfa/iprscan/) [60] and MOTIFSCAN (myhits.isb-sib.ch/cgi-bin/motif_scan) [61] databases. The *Dof* protein sequences were then subjected to NUCPRED server (www.sbc.su.se/~maccallr/nucpred/) [62] for the identification of nuclear localization signal (NLS). The expasy server (http://web.expasy.org/compute_pi/) [63] was used to compute pI and molecular weight of identified *CcDof* proteins.

Protein alignment and phylogenetic analysis

Multiple sequence alignment of the full-length deduced amino-acid sequences of *Dof* proteins were performed by Clustal X2.0.10 [64]. A phylogenetic tree was constructed using the MEGA 5.2.2 program (www.megasoftware.net/mega.php) [65] and Neighbor-joining (NJ) method [66] with Poisson correction, random seed of phylogeny test and pairwise deletion option parameters enabled. The reliability of the trees obtained was tested using a bootstrap test with 1,000 replicates.

Mapping of *CcDof* genes on pigeonpea chromosomes and its intron/exon gene structure prediction

The chromosomal locations of pigeonpea *Dofs* were mapped to the duplicated blocks using the CViT (Chromosome Visualization Tool) genome search and synteny viewer at the Legume Information System (<http://comparative->

legumes.org) [67, 68]. The deduced nucleotide sequences of all *CcDofs* were used to search against the pigeonpea genome and the results were displayed manually.

Identification of conserved motif

The deduced protein sequences of the 38 *CcDof* genes were analyzed by means of the MEME (Multiple EM for Motif Elicitation) program software version 4.9.0 [69] for motif analysis. To identify conserved motifs in these sequences, the selection of maximum number of motif was set to 25 with minimum and maximum width of 6 and 55 amino-acid respectively. The distribution of one single motif was considered as any number of repetitions while the other factors were of default selections.

Cis-regulatory element analysis

For promoter analysis, 1,000-bp sequences upstream from the TSS (transcription start site) of the putative *CcDofs* were retrieved. These sequences were then subjected to CARE program (<http://bioinformatics.psb.ugent.be/webtool/plantcare/html/>) of PlantCARE database [70] for the identification of putative *cis*-regulatory elements.

Inference of duplication time

The K-estimator 6.1 program was used to estimate the K_a and K_s values of paralogous genes. Estimates of evolutionary rates are useful for explaining patterns of macro-evolution because K_s can be used as a proxy for time when estimating dates of segmental duplication events. The K_s value was calculated for each of the gene-pair and then used to calculate the approximate date of the duplication event ($T = K_s/2\lambda$), assuming the mean value of clock-like rates (λ) of synonymous substitution (2.0×10^{-9}) [71, 72].

Functional divergence analysis

To estimate levels of functional divergence and to predict amino-acid residues responsible for functional differences in Dof groups, the coefficients of type-I functional divergence were calculated using the method suggested by Gu [73, 74]. Further analysis were carried out using DIVERGE 2.0 [73, 74], which uses maximum likelihood procedures to estimate significant changes in site-specific shifts in evolutionary rates after gene duplication or speciation. The advantage of this method is that it uses amino-acid sequences and thus not sensitive to saturation of synonymous sites. The Type-I functional divergence designates amino-acid configurations that are highly conserved in gene 1 but highly variable in gene 2, or vice versa, implying that these residues have experienced altered

functional constraints [73]. The coefficients of functional divergence significantly greater than 0 indicate site-specific altered selective constraints or radical shifts of amino-acid physiochemical properties after gene duplication.

Prediction of ancestral protein sequences

FAST-ML 2.02 (www.tau.ac.il/~talp/supplementary/fastml/fastml.2.02) [75] was used to predict ancestral sequences from phylogenetic trees and the corresponding multiple sequence alignments. The input tree was unrooted and it was rooted automatically by midpoint rooting in FAST-ML Ancestral Sequence Prediction analysis.

Results and discussion

Identification of Dof genes

A total of 38 non-redundant *Dof* transcription factor genes were annotated from the whole genome sequence of pigeonpea by performing a BLAST search against the genome using amino-acid sequence of the conserved Dof domain. The presence of a typical Dof DNA binding domain of 52 amino-acid residues spanning a single C2/C2 zinc finger structure revealed the true identity of the annotated *Dof* genes, which were tentatively named as *CcDof1* to *CcDof38*.

The protein functional analysis of these *CcDof* proteins using INTERPROSCAN has shown identity with InterPro accession number IPR003851. These proteins also showed similarity with signature accession namely PD007478 (ProDom database), PF02701 (Pfam database), PS01361 (PROSITE patterns database) and PS50884 (PROSITE profiles database) confirming their identity to Dof-like proteins. The deduced protein sequences of *CcDof* genes were then subjected to motif scan database. Motif scan integrated with PeroxiBase profiles, PROSITE patterns, PROSITE profiles, HAMAP profiles, Pfam HMMs (local models), Pfam HMMs (global models) and Peroxi-Base profile databases revealed the presence of proline, serine, threonine, glutamine, asparagine, glycine and histidine rich regions in different *CcDof* genes.

A total of 13 *CcDof* proteins namely *CcDof* 2, *CcDof* 4, *CcDof* 10, *CcDof* 13, *CcDof* 14, *CcDof* 16, *CcDof* 18, *CcDof* 28, *CcDof* 32, *CcDof* 33, *CcDof* 34, *CcDof* 35 and *CcDof* 37 showed nuclear localization signal (NLS) when subjected to NUCPRED software. The NLS signal for *CcDof* 2, *CcDof* 13, *CcDof* 14, *CcDof* 16, *CcDof* 18, *CcDof* 28, *CcDof* 33, *CcDof* 34, *CcDof* 35 and *CcDof* 37 was found to be AGRRK while *CcDof* 4 and *CcDof* 10 showed GAGRRKTKN and AGKRRK, respectively. In case of *CcDof* 32 PKRRR and GSSG was observed as NLS signal.

The Dof domain of pigeonpea revealed highly-conserved sequences and 26 out of 52 amino-acid were found

to be 100 % conserved in all the sequences. These sequences also include conserved cysteine residues that coordinate with zinc ion. Other conserved residues observed were Asn¹⁰, Tyr¹⁶, Asn²⁰, His²⁶, Lys²⁹, Arg³², Tyr³⁴, Met⁴¹, Asn⁴³, Val⁴⁴, Pro⁴⁵, Val⁴⁶, Gly⁴⁸, Arg⁵¹ and Lys⁵² as shown in Fig. 1. The other 11 amino-acid residues were found to be different in all the CcDof proteins.

The identified *CcDof* genes encodes peptides ranging from 153 to 544 aa with the pI value varying from 5.27 to 10.18, and the molecular weight ranging from 17.1 to 59.3 kD revealed from ExPASy server (http://web.expasy.org/compute_pi/). Most members within the individual sub-groups as shown in phylogenetic tree (Fig. 2a) revealed similar amino-acid sequence lengths, isoelectric point and molecular weights (Table 1). The corresponding orthologs of *Dof* genes of Arabidopsis from the annotated *Dof* genes of pigeonpea is also mentioned in Table 1.

Gene structure prediction

The exon–intron organization has been used as supporting evidence for evolutionary relationships among genes or

organisms [76]. To get an insight about the possible mechanisms of structural evolution of *Dof* genes in pigeonpea, comparison of the exon–intron structures of 38 annotated *Dof* genes was performed. The distribution and position of introns within each of the *CcDof* gene family is shown in Fig. 3. Unlike rice, Arabidopsis and soybean gene structures, the number of introns ranged from 0 to 4 in *CcDof* genes. A total of 17 out of 38 *CcDof* genes were found to be intronless. A total of 17 out of 36 *Dof* gene models in rice [8], 21 out of 43 *Dof* gene models in Arabidopsis [8] and 35 out of 78 *Dof* gene models in soybean are intronless [12]. The *CcDof25* has a maximum of four introns, while *CcDof14* has three introns. Two of the *CcDof* genes (i.e. *CcDof6* and *CcDof18*) have two introns, while a total of 17 *CcDof* genes possess only one intron. The members of *Dof* genes representing the same group in phylogenetic tree revealed similar coding sequences and exon–intron structures, strongly supporting their close evolutionary relationship. The positions and distribution of some introns of *CcDof* genes are conserved in genes of same cluster, while others were found to be group-specific. All the members of *CcDof* genes

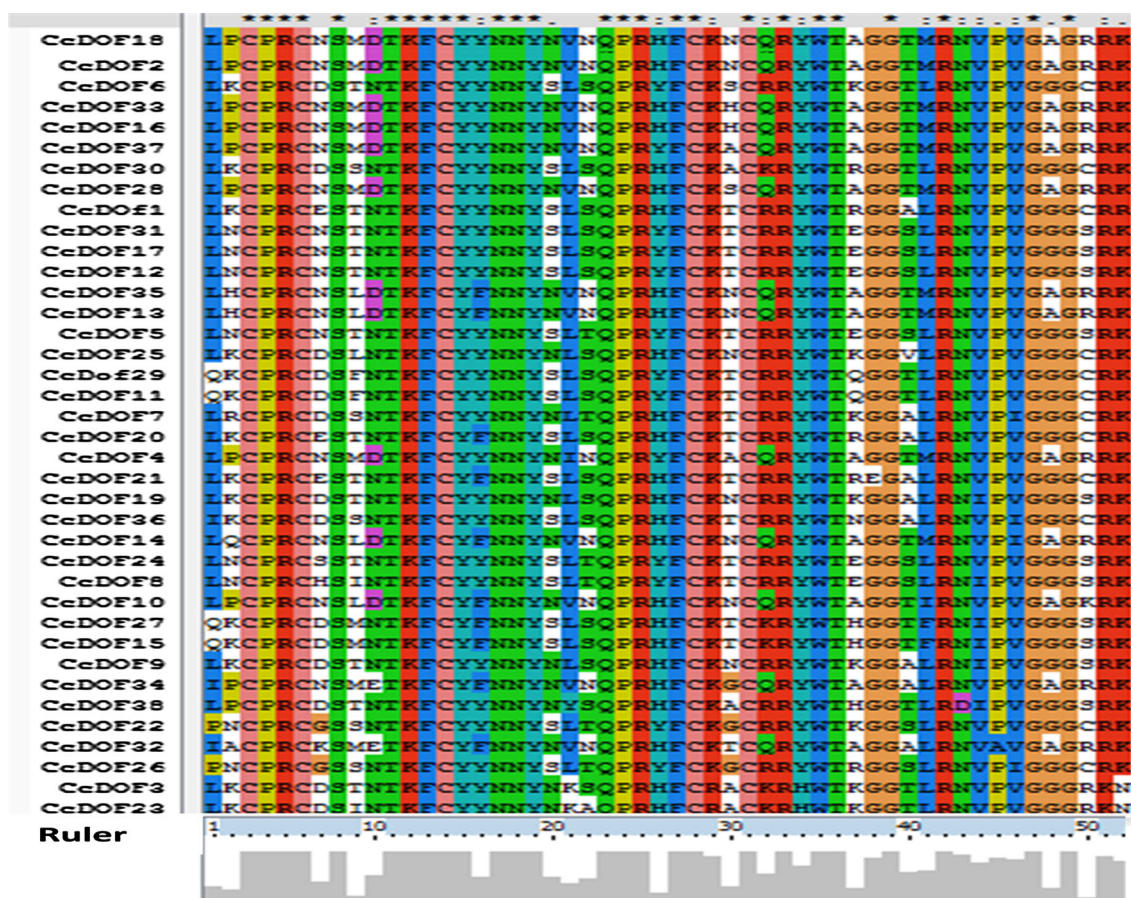


Fig. 1 Multiple sequence alignment of the Dof DNA binding domain among 38 CcDof proteins. Strongly conserved residues are indicated above the alignment with asterisks, colons indicates the residues

variation occurring within the strongly conserved groups and dots indicating the residues variation occurring within weaker conserved residue groups

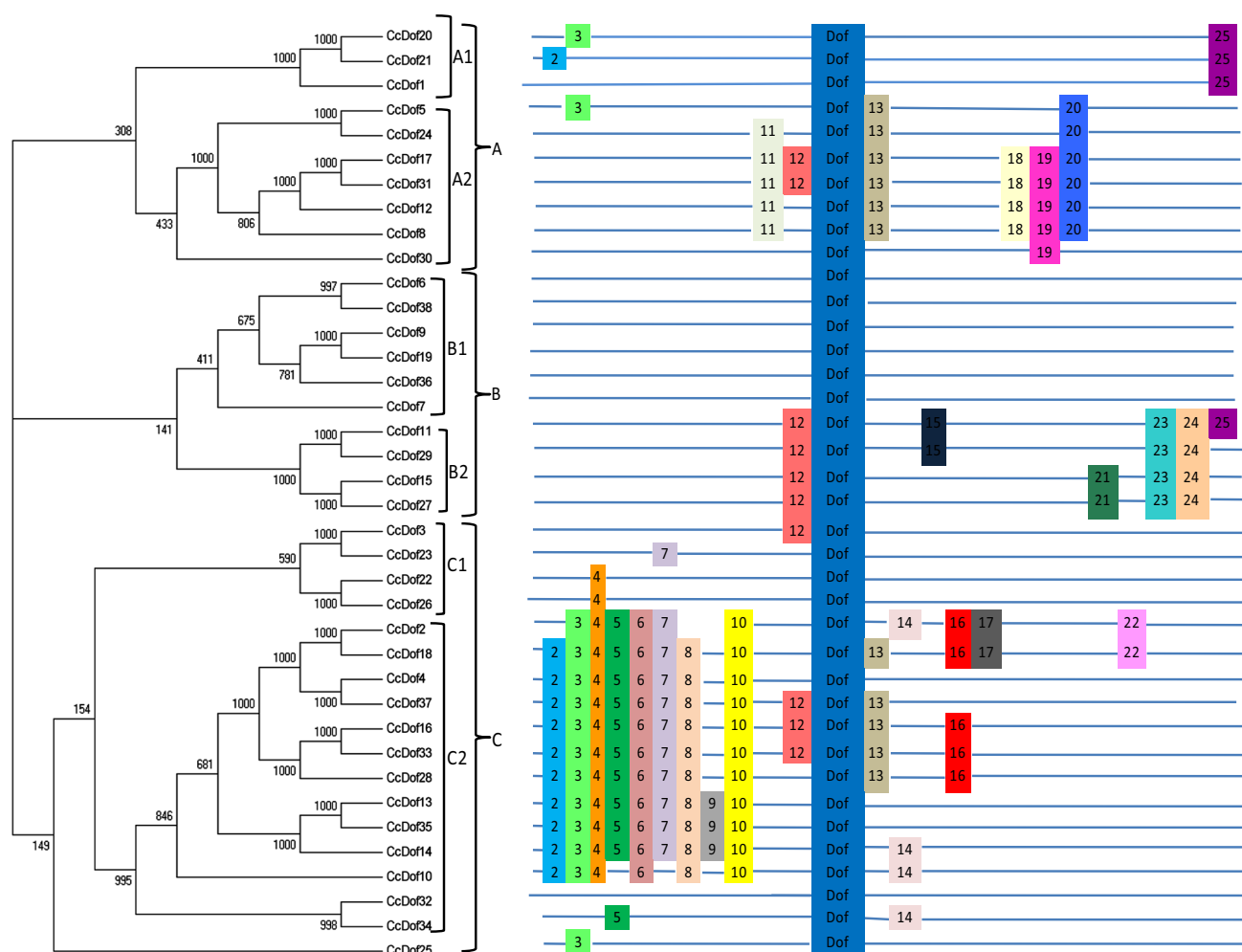


Fig. 2 **a** Phylogenetic tree constructed by NJ method using the MEGA5.2.2 program based on protein sequences. **b** Distribution of 25 motifs among *CcDof* protein using MEME ver. 4.9.0. Motif 1 is the conserved Dof domain indicated with dark blue bar

representing subcluster A1 were having one intron and C1 represented genes without introns. Introns are considered to be important and necessary subassemblies of eukaryotic genes as their loss or gain generates structural diversity and complexity which leads to the evolution of multiple gene families. Intron's potential functions include serving as a source of regulatory elements, role in exon shuffling and alternative splicing, and as a source of signals for exporting mRNA from the nucleus [77, 78]. It has also been suggested that introns can increase intragenic recombination and moderate the evolutionary rate of genes [79]. In addition, growing evidence has shown a functional link between introns and gene expression [80].

Chromosomal location

To determine the genomic distribution of the *CcDof* genes, the chromosomal locations of these genes were mapped to the duplicated blocks using the CViT (Chromosome

Visualization Tool) genome search and synteny viewer at the Legume Information System (<http://comparative-legumes.org>) [67, 68]. The result revealed the distribution of these *CcDof* genes among most of the eleven chromosomes except chromosome number 2, 5 and 9. A maximum of 11 and 9 *Dof* genes were observed in chromosome 11 and 1, respectively, while only single *Dof* gene was found on chromosome 3. The distribution of *Dof* genes among different chromosomes are shown in Fig. 4. The distribution of *Dof* genes of rice and *Arabidopsis* among the chromosome has been reported [8]. In case of rice, the 30 *Dof* genes were distributed on 11 out of 12 chromosomes, with a maximum of 6 *Dof* genes observed on chromosomes 1 and 3. Similarly in case of *Arabidopsis*, the 36 *Dof* genes were distributed among all the 5 chromosomes with a maximum of 9 *Dof* genes on chromosome 1. In soybean, chromosome 13 contained the largest number of 11 *Dof* genes while no *Dof* gene was found on chromosome 14 [12].

Table 1 List of *Dof* genes identified in pigeonpea with their corresponding proteins, CDS and protein length along with orthologs identified from *Arabidopsis thaliana* *Dof* gene family

Gene	Source Accession No.	Ortholog locus in Arabidopsis	Chromosome number and location	Gene structure	CDS length (bp)	Protein length (aa)	pI	Mw (kD)
<i>CcDof1</i>	AFSP01002933.1	At2G37590.1	10 (2105415-2105545 bp)	One intron	1122	373	9.46	40.1
<i>CcDof2</i>	AFSP01004960.1	At5G39660.2	7(15118662-15120946 bp)	Four intron	1635	544	6.83	59.3
<i>CcDof3</i>	AFSP01004978.1	At1G51700.1	8(17704698-17705016 bp)	Intronless	924	307	8.62	33.7
<i>CcDof4</i>	AFSP01006005.1	At3G47500.1	11(18816432-18820434 bp)	One intron	1332	443	6.29	48.2
<i>CcDof5</i>	AFSP01007923.1	At3G61850.1	11(8951627-8954735 bp)	Intronless	831	276	9.04	30.1
<i>CcDof6</i>	AFSP01008739.1	At2G28510.1	4(5526255-5528097 bp)	Two introns	1239	412	7.97	31.9
<i>CcDof7</i>	AFSP01009569.1	At1G51700.1	6(16571571-16571742 bp)	Intronless	987	328	8.80	35.0
<i>CcDof8</i>	AFSP01010784.1	At2G46590.2	11(8952832-8952988 bp)	One intron	888	295	8.93	32.2
<i>CcDof9</i>	AFSP01011066.1	At1G51700.1	11(4230484-4232004 bp)	Intronless	621	206	8.77	21.8
<i>CcDof10</i>	AFSP01011681.1	At5G39660.2	7 (15120466-15120605 bp)	One intron	1425	474	6.09	52.7
<i>CcDof11</i>	AFSP01018685.1	At5G60200.1	1(7838244-7838385 bp)	Intronless	948	315	9.37	33.8
<i>CcDof12</i>	AFSP01035278.1	At3G61850.3	11(42579241-42580117 bp)	One intron	864	287	8.88	31.4
<i>CcDof13</i>	AFSP01036634.1	At5G39660.2	7(7004757-7005222 bp)	One intron	1440	479	5.93	52.3
<i>CcDof14</i>	AFSP01039940.1	At5G39660.2	7(7000086-7006313 bp)	Three introns	1623	540	7.95	59.0
<i>CcDof15</i>	AFSP01042849.1	At3G55370.1	1(7839007-7837991 bp)	Intronless	849	282	10.18	30.3
<i>CcDof16</i>	AFSP01053573.1	At5G39660.2	1(11075668-11075855 bp)	One intron	1383	460	6.70	50.2
<i>CcDof17</i>	AFSP01057769.1	At3G61850.3	11(42579857-42580062 bp)	One intron	870	289	7.61	31.8
<i>CcDof18</i>	AGCT01000940.1	At5G39660.2	7(15118662-15120946 bp)	Two introns	1584	527	6.58	57.3
<i>CcDof19</i>	AGCT01001242.1	At1G51700.1	10(310410-312780 bp)	Intronless	639	212	8.77	22.9
<i>CcDof20</i>	AGCT01003200.1	At3G55370.1	4(3031047-3033532 bp)	One intron	924	307	9.53	33.1
<i>CcDof21</i>	AGCT01003200.1	At3G55370.1	4(3015147-3016581 bp)	One intron	912	303	9.30	32.9
<i>CcDof22</i>	AGCT01004026.1	At3G52440.1	6(2975753-2977677 bp)	Intronless	867	288	5.27	31.9
<i>CcDof23</i>	AGCT01006387.1	At1G51700.1	8(17703797-17706646 bp)	Intronless	849	282	8.65	31.4
<i>CcDof24</i>	AGCT01008337.1	At3G61850.4	3(28698326-28700377 bp)	Intronless	744	247	8.51	27.1
<i>CcDof25</i>	AGCT01010495.1	At3G52440.1	11(32078732-32079844 bp)	Intronless	735	244	6.80	27.0
<i>CcDof26</i>	AGCT01010642.1	At1G28310.2	11(34586100-34587497 bp)	Intronless	765	254	5.64	28.8
<i>CcDof27</i>	AGCT01012076.1	At3G55370.1	1(7837991-7839007 bp)	Intronless	849	282	10.18	30.3
<i>CcDof28</i>	AGCT01012296.1	At5G39660.2	1(11074602-11077174 bp)	One intron	1377	458	8.00	49.3
<i>CcDof29</i>	AGCT01022941.1	At5G60200.1	1(7838244-7838385 bp)	Intronless	948	315	9.37	33.8
<i>CcDof30</i>	AGCT01022943.1	At5G60200.1	1(7806607-7806802 bp)	One intron	1089	362	8.78	38.9
<i>CcDof31</i>	AGCT01038186.1	At3G61850.3	11(42579857-42580062 bp)	One intron	870	289	7.61	31.8
<i>CcDof32</i>	AGCT01041956.1	At2G34140.1	1(11075728-11075821 bp)	One intron	462	153	9.65	17.1
<i>CcDof33</i>	AGCT01042559.1	At5G39660.2	1(11075668-11075855 bp)	One intron	1383	460	6.70	50.2
<i>CcDof34</i>	AGCT01046526.1	At2G34140.1	7(14616628-14616823 bp)	Intronless	486	161	9.36	18.1
<i>CcDof35</i>	AGCT01052531.1	At5G39660.2	7(7004757-7005222 bp)	One intron	1440	479	5.93	52.3
<i>CcDof36</i>	AGCT01059318.1	At2G375900.1	6(13193931-13194066 bp)	Intronless	1053	350	8.72	37.2

Table 1 continued

Gene	Source Accession No.	Ortholog locus in Arabidopsis	Chromosome number and location	Gene structure	CDS length (bp)	Protein length (aa)	pI	Mw (kD)
<i>CcDof37</i>	AGCT01059868.1	At3G47500.1	11(18818151-18818356 bp)	One intron	1329	442	5.67	48.4
<i>CcDof38</i>	AGCT01071779.1	At3G50410.1	11(32080779-32080928 bp)	Intronless	699	232	8.72	23.9

Phylogenetic relationship and motif analysis

To reveal the functional similarity among 38 annotated *Dof* genes, phylogenetic analyses of the *Dof* protein sequences based on Maximum likelihood (ML) method and Neighbour joining (NJ) method using MEGA 5.2.2 was performed [65]. The tree topology assessed by the ML method was substantially similar to the tree constructed by NJ method. From the phylogenetic tree constructed, the *CcDof* proteins were grouped into three major clusters designated as A, B and C (Fig. 2a). The major cluster A, B and C were further divided into sub-clusters namely A1 and A2, B1 and B2, C1 and C2, respectively. A very high bootstrap value suggested common origin for *Dof* genes of each subgroup. Inspection of the phylogenetic tree topology revealed several pairs of *Dof* proteins with a high degree of homology in the terminal nodes of each subgroup, suggesting that they are putative paralogous pairs (Fig. 2a). A total of 14 pairs (*CcDof20/CcDof21*, *CcDof5/CcDof24*, *CcDof17/CcDof31*, *CcDof6/CcDof38*, *CcDof11/CcDof29*, *CcDof9/CcDof19*, *CcDof15/CcDof27*, *CcDof3/CcDof23*, *CcDof22/CcDof26*, *CcDof2/CcDof18*, *CcDof4/CcDof37*, *CcDof16/CcDof33*, *CcDof13/CcDof35* and *CcDof32/CcDof34*) of putative paralogous *Dof* proteins were identified. These many pairs of paralogous genes supported the hypothesis that they might have been evolved from a genome duplication event.

A total of 25 conserved motifs were analyzed by means of MEME software. The schematic distribution of these 25 motifs among *CcDof* proteins are shown in Fig. 2b. The identified multilevel consensus sequence for the motifs is shown in Table 2. The closely related *Dof* members as represented by different clusters in the phylogenetic tree revealed common motif distribution, suggesting functional similarities among the *Dof* proteins within the same sub-clusters (Fig. 2a). For instance, motif 25 for A1 sub-cluster, motifs 11, 13, 18, 19 and 20 for A2 sub-cluster, motifs 12, 23 and 24 for B2 sub-cluster, motifs 2, 3, 5, 6, 7, 8 and 10 for C2 sub-cluster. The motifs distribution also highlights that the *Dof* genes are supposed to be conserved during evolution.

Cis-regulatory element analysis

The *cis*-regulatory elements, which are the binding sites for transcription factors located in the promoter regions of genes, are the functional elements that determine the spatial and temporal transcriptional activity. A large number of *cis*-elements have been identified by promoter analysis which serves as an important molecular switch associated with the transcriptional regulation of a dynamic network of gene activities controlling multifarious biological processes. To get an insight about the putative functions of

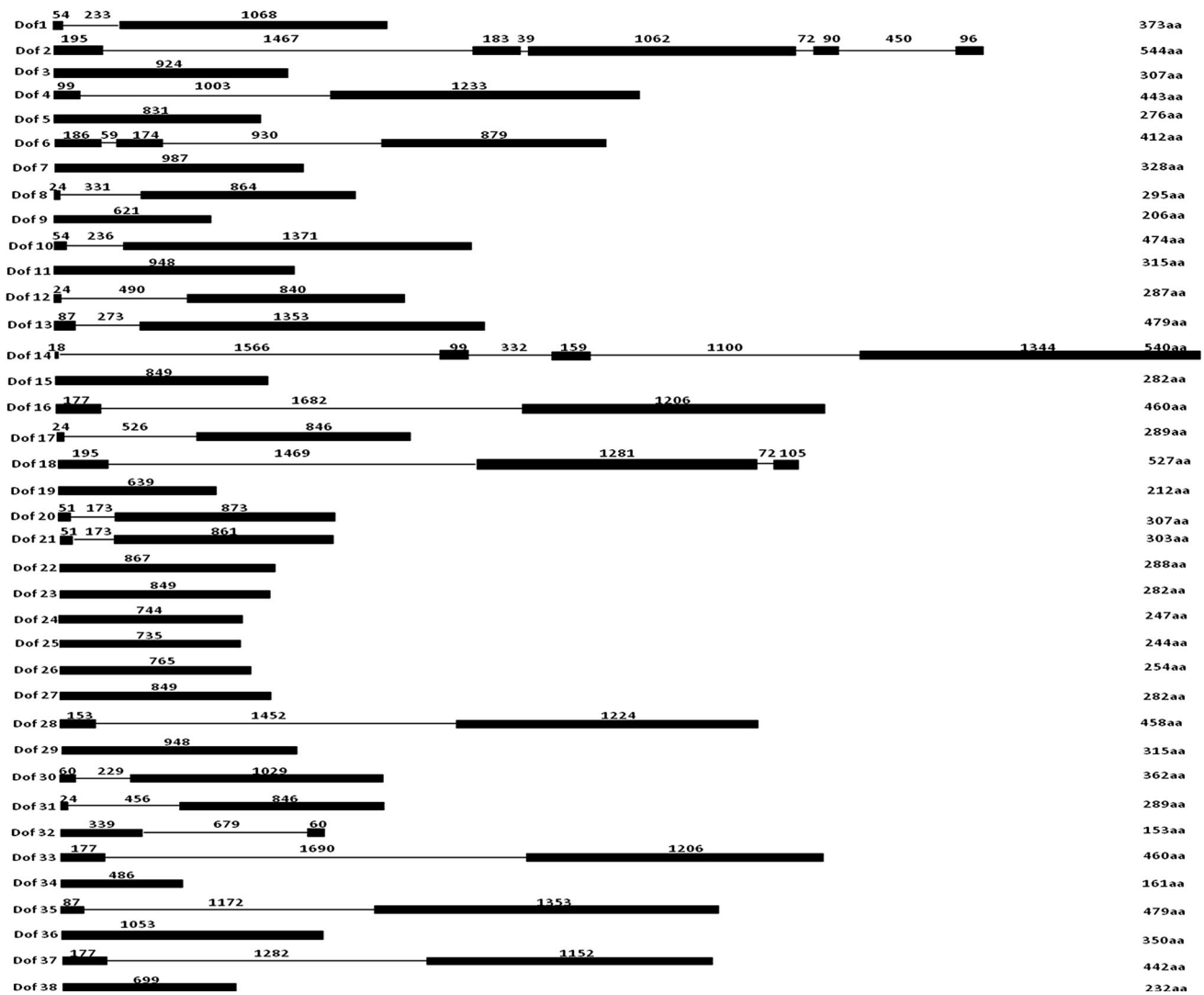


Fig. 3 Intron/exon structure of *CcDof* genes. Black bars represent the exon while the lines represent the intron. The number denoted above the lines indicate the length of exons and introns in base pairs. The column at the right hand side represents the length of Dof proteins in amino-acids

Dof TF, 1,000 bp upstream sequence were retrieved from putative transcription start site from pigeonpea genomic sequence database. A large number of *cis*-regulatory elements were observed when subjected to PlantCARE database. The *cis*-regulatory elements associated with several important physiological phenomena namely responses to light, hormone, stresses, endosperm and meristem specific gene expression were analyzed extensively as shown in Fig. 5a. The predominance of many light responsive and stress responsive elements strongly indicates the possibility of *CcDof* genes association with regulation of photoperiodic control of flowering and also involvement in abiotic and biotic stress (Fig. 5b). The role of Dof transcription factors in biotic stress tolerance have been investigated in barley seeds and in *Arabidopsis* based on its interaction with cystatin gene and *OCS* (octopine synthase) gene

respectively [6, 81, 82]. The Dof proteins are also involved in light responses to plants. OBP3, COG1 are *Arabidopsis* Dof proteins that regulate phytochrome signaling [83, 84]. In *Arabidopsis* several Dof transcription factors, namely *CDF1*, *CDF2*, *CDF3*, *CDF5* and *OsDof12* of rice are reported to be essential for photoperiodic control of flowering [85–87]. A maximum number of 13 light responsive elements were observed in the promoter region of *CcDof* 11 gene. The presence of GCN4-motif, O2-site, Skn-1, RY-element and Sp1 element confers endosperm specific gene expression. Similarly presence of *cis*-regulatory elements like ABRE, P-box, ERE, AuxRR-core, P-box, TATC-box, GARE-motif, TGA-element reveals its association with hormone specific expression. The involvement of *CcDof* gene in regulation of meristem specific expression and having protein binding site was also observed. The list of

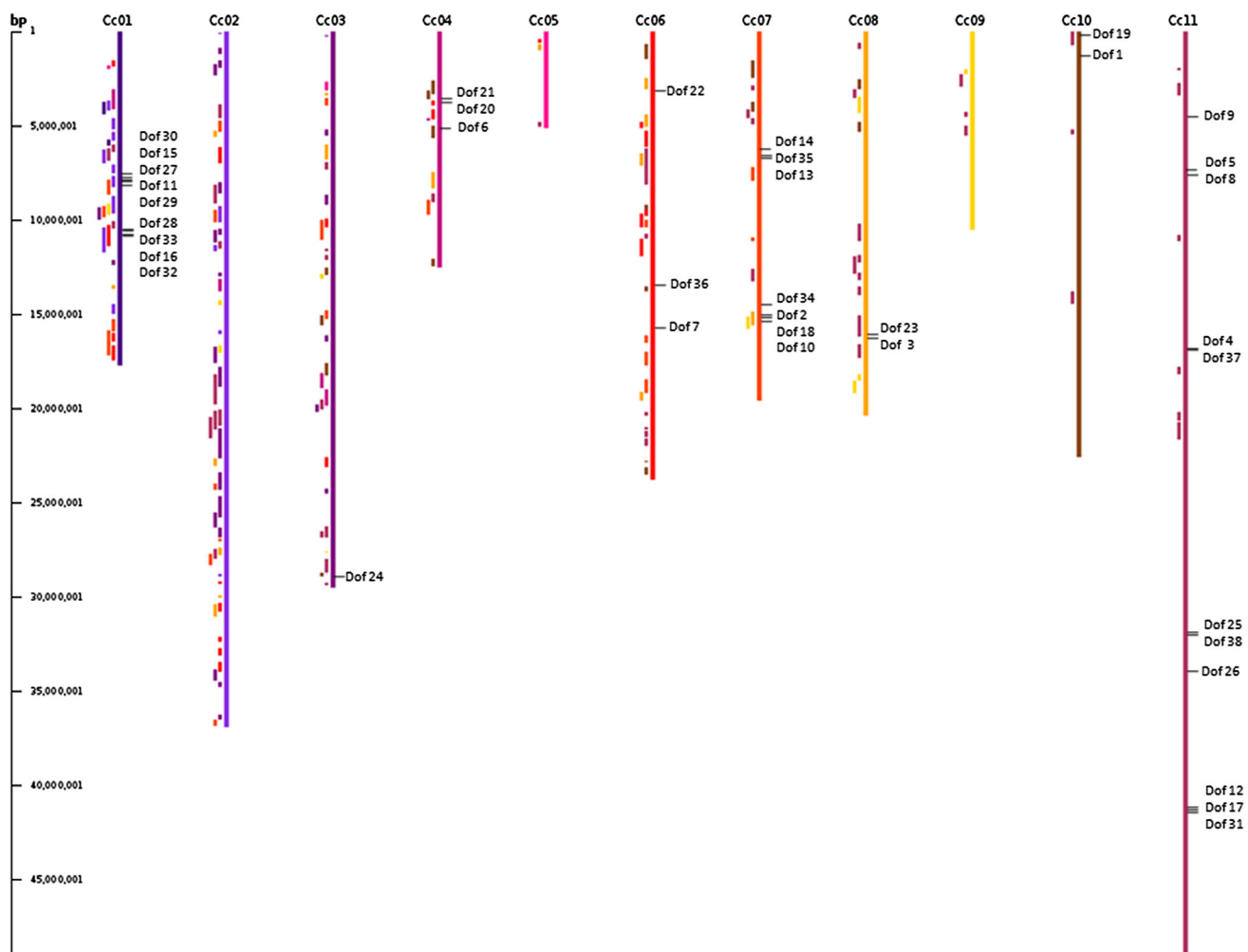


Fig. 4 Chromosomal locations, region duplications and predicted clusters for *CcDof* genes of pigeonpea. The colored blocks to the left of each chromosome show duplications with chromosomes of the same color. Chromosomal numbers are indicated at the top of each bar

identified *cis*-regulatory elements among the 38 *CcDof* genes is shown in Supplementary Table S-1 and the putative functions of these elements are represented in SupplementaryTable S-2.

Gene duplication and ancestral protein sequences

Based on phylogenetic tree, a total of 14 pairs of paralogous genes were identified in the terminal nodes as supported by strong bootstrap values. The location of these genes was either on the same chromosome or on different chromosomes, suggesting that tandem duplication and segmental duplication together might have contributed to the expansion of the *Dof* gene family in pigeonpea. As shown in Fig. 4, five pairs of the paralogous genes, randomly scattered throughout the genome, were supposed to be resulted from putative segmental duplication events, while nine pairs of paralogous genes co-located at the chromosome level were thought to be resulted from tandem

duplication (Table 3). The genes duplication pattern revealed tandem duplication to be predominant over segmental duplication events in case of *CcDof* genes. An attempt was also made to predict the immediate ancestral protein sequences of 14 pairs of paralogs and the identified protein motif structure of the duplicated gene (Fig. 6 and Table 4). The K_a/K_s ratio is a measure for selection pressure on amino-acid substitutions and the value greater than 1 suggests purifying selection [88, 89]. The amino-acid in a protein sequence are expected to be under different selective pressures and are indicated by different underlying K_a/K_s ratios and detection of selective pressure can indicate selective advantages for altered amino-acid sequences in the residues. These selective advantages are essential for understanding the functional residues and functional protein shifts [90]. The ratios of non-synonymous (K_a) versus synonymous (K_s) mutations have been used to analyze positive and negative selection of specific amino-acid sites within full-length *Dof* protein sequences among different

Table 2 Multilevel consensus sequences for the MEME defined 25 different motifs observed among 38 identified Dof proteins of pigeonpea. The mean amino-acids are shown without square brackets

Motif	No. of a.a.	No. of sites	E value	Multilevel consensus sequence
1	55	38	1.4e-1845	L[KPN]CPRC[ND]S[TM][ND]TKFCY[YF]NNY [NS][LV][SN]QPR[HY]FCK[TN]C[RQ]RYWT[AK]GG[TAS][LM] RNVPGV[GA]G[RCS]RKNK[RNK]
2	29	11	5.3e-149	[EN]K[SC][VL][LW]VPKTLRIDDP[NS] EA[AS]KS[SP]IWSLGI
3	29	11	6.1e-111	[EG]K[ND][HR]VV[EG]AS[PQ][VI]L[QE] ANPAA[LI]SR[SA][LH]AF[HQ]E[QRS]T
4	28	10	1.1e-110	[KTL][DCH][NT][GS][TA]VL[TKS]FG[PLST]D [AST]P[LI]C[DE]SM[AE]S[VM]L[NS][LI][AGK][DE][KP]
5	24	11	9.3e-78	[MS]SENKDP[AG]IKLFGR [TK][PS][LV]P[EG][CN]QI[AP]
6	29	11	1.1e-071	P[SPT]SST[SP][NA]S[PCT][CNS]SG [PN]NSPTLGKH[ST]R[DE][GD][DN]IL[KF]
7	15	10	1.5e-052	[SL]AS[HQ]YR[HQ][TIM] [VIL][SPA][ES][ADT][LGA][IQ]
8	21	10	1.2e-044	[EG][DK][NEQ][MQS][DEK][ENT][ST] [SI][DLNT][SD]Q[DE]K[ATI][LPF]KKPDK[IV]
9	37	3	1.0e-033	[HP]Y[SN][AT][DG]PASMQWCPTPMVA [VI][QP][TG][IM]CPPS[FI]PLQ[FL]VP[GP]SYW
10	21	11	4.0e-036	[EN][HE][GN]xSPQ[LV][PS][CFY] F[PS][GV][PT]PWPYPWN
11	48	3	1.2e-038	Q[TD]PKIHQGQDLNLAYPPVEDY [SA]T[IV]SKFIEVPY[NS][NT][TE][EL]D[NK][NG][KL][AQ][EN] [TQ][EN][TP][HT][HS][QA]
12	29	9	2.0e-033	[MGP][EI][QH]E[GA][EG][RDG][DART] [EKNR][EIK][HLNP][FPRS][QPR][AKQV][EQV][DLQ][RLT][AKSV][ASTV] [QK][QDE][QHIM][EMQT][QDK][ANPQ][AGPS][EDHQ][ADFQ][QE]
13	15	6	1.1e-031	[DN]STGYW[NT]GM[LI]G[GE]GSW
14	15	8	2.8e-021	[DEH]S[ILV][SNP]GG[GS] [LM]FKAF[PAQ][SP]K
15	55	2	1.6e-019	RAIGPIRPPQLYQMGSGEREFVSLYEQGL VNPSISSSMANTNTNISGVSQHDWSR
16	11	5	9.4e-019	FY[PT][AT][PA][AP][YF]WGC[MA]
17	55	2	1.2e-017	RDAPVDLDKDTLGEKPTDNKKEDGAPT QSSEEVTPDAASGTCEESITLSTERE
18	21	4	2.3e-016	[MV][EV][GT][NS][SKT][KAC]P[PK][IGM] [MLV]E[KR][RK]ARPQK[DE]QA
19	13	4	4.2e-015	MDT[AT][QH]W[AP]Q[GE][IM][GV]V[VK]
20	9	10	4.2e-017	[GL]FP[FLM][QEG][DE][FILV]K[PQ]
21	53	2	2.1e-017	RFHHMGRVESLYPTQQLVIPSNVGASATG SHTTQLPQSLIVSNNNATNASLW
22	55	2	2.7e-017	VHNYTRNGFHKPEELRISVPLASGEKGDDQ SNKSSITSTKSMEGTSTNGPQEAM
23	27	4	3.5e-015	[AV][GV]A[GS][PV][GS][MV]G[PV][FG][VY] [GH][GS][GS]G[FY]LSSLA[AT][LM]HSLN
24	29	4	1.6e-014	[NR][NT]S[AD][QS][NS][NS][AV][GK][AV] G[GS]SS[FL][IV][PQ]N[HQ]W[LP][HN]LPGY[AG]PP
25	55	2	3.3e-014	MVFSSIPVFVDPPNWQQPN[HN]PQANGSDN [ST]QLLPPLPPQ[HP]HVG[FS]VGSIRPGSM

groups. The result indicated the variability in the Ka/Ks ratios of the sequences from different Dof groups as shown in Table 3. However, despite the differences in Ka/Ks

values, most of the estimated Ka/Ks values were substantially lower or equal to 1, suggesting that the Dof sequences within each of the groups are under strong

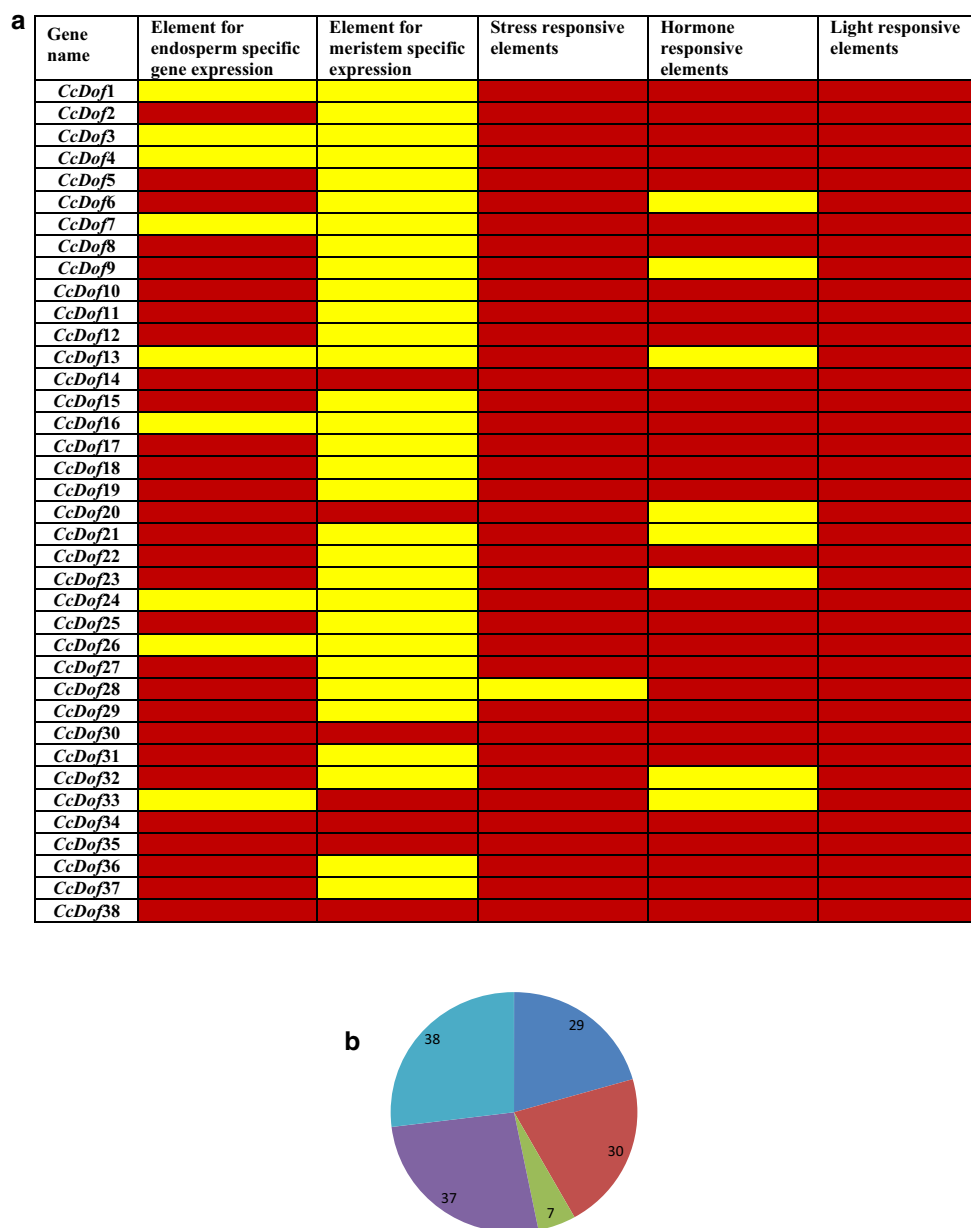


Fig. 5 a Presence (red) and absence (yellow) of *Cis*-regulatory elements in *CcDof* gene family, associated with 5 important physiological phenomena. **b** Frequency of *Cis*-regulatory elements associated with gene specific expression namely, Endosperm specific

gene expression (yellow); Hormone responsive element (red); Meristem specific expression (green); Stress responsive elements (purple) and Light responsive elements (cyan)

purifying selection pressure and that positive selection might have acted only on a few sites during the evolutionary process.

The approximate dates of duplication events were also calculated using Ks. The segmental duplications of the *Dof* genes in pigeonpea is supposed to be originated from 35.9 Mya (million years ago, Ks = 0.1438) to 261.5 Mya (Ks = 1.046), with the mean of 136.18 Mya (Ks = 0.544); the Ks of tandem duplication of the *Dof* genes ranged from

0.099 to 2.077, dating the duplication event between 24.7–519.2 Mya. The date for duplication event for five pairs of paralogous genes could not be calculated, as these pair of genes showed difference only in their intron sequence (i.e. *CcDof17/CcDof31*, *CcDof11/CcDof29*, *CcDof15/CcDof27*, *CcDof16/CcDof33* and *CcDof13/CcDof35*). The Ka/Ks ratios of three segmental duplication pairs and all nine tandem duplication pairs were more than 0.3, suggesting that significant functional divergence might

Table 3 The gene duplication time identified in different paralogous pairs of *CcDof* gene families. The K_a represents the number of non-synonymous substitution per non-synonymous site while K_s is thenumber of synonymous substitution per synonymous site and K_a/K_s represents the ratios of non-synonymous (K_a) versus synonymous (K_s) mutations

Paralogous pairs	Location on chromosome	Duplication event	K_a	K_s	K_a/K_s	Date (million years ago)
CcDof20/CcDof21	Chr4/Chr4	Tandem	0.06504	0.11407	0.57017	28.5
CcDof5/CcDof24	Chr 11/Chr3	Segmental	0.12326	0.43927	0.28060	109.8
CcDof17/CcDof31	Chr11/Chr11	Tandem	0	0	1	–
CcDof6/CcDof38	Chr4/Chr11	Segmental	1.333	1.046	1.27437	261.5
CcDof11/CcDof29	Chr1/Chr1	Tandem	0	0	1	–
CcDof9/CcDof19	Chr 10/Chr11	Segmental	0.16496	0.55750	0.29589	139.3
CcDof15/CcDof27	Chr1/Chr1	Tandem	0	0	1	–
CcDof3/CcDof23	Chr8/Chr8	Tandem	0.13208	0.40264	0.32803	100.6
CcDof22/CcDof26	Chr6/Chr11	Segmental	0.23021	0.53785	0.42801	134.4
CcDof2/CcDof18	Chr7/Chr7	Tandem	1.079	0.099	10.89	24.7
CcDof4/CcDof37	Chr11/Chr11	Tandem	2.399	2.077	1.15503	519.2
CcDof16/CcDof33	Chr1/Chr1	Tandem	0	0	1	–
CcDof13/CcDof35	Chr7/Chr7	Tandem	0	0	1	–
CcDof32/CcDof34	Chr1/Chr7	Segmental	0.4721	0.1438	3.28303	35.9

have occurred after duplication events. All these data suggest that the pigeonpea Dof family segmental/tandem expansion could be dated to recent duplication events, and that these *Dof* genes retained their function after duplication.

Functional divergence

An attempt was made to find out whether amino-acid substitution causes adaptive functional diversification by performing Type-I functional divergence between gene clusters of the pigeonpea Dof family by posterior analysis using the program DIVERGE [73, 74], which evaluates shifts in rates of evolution and altered amino-acid properties. The value of the coefficient of functional divergence (Θ) estimated for ClusterA/ClusterB, ClusterB/ClusterC and ClusterC/ClusterA were -0.012275 , -0.010152 and 0.092603 , respectively. The coefficient of all functional divergence (Θ) values between these clusters is less than 1 indicating significant site-specific selective constraints for most of the members of the pigeonpea Dof family. This might have led to cluster specific functional evolution after diversification [91, 92]. A higher Theta values (Θ) existed in Cluster C/Cluster A (0.092603), indicating a higher evolutionary rate or site-specific selective relaxation between them. Therefore, because of the different evolutionary rates predicted at some amino-acid sites, the *Dof* genes might be significantly divergent from each other in their functions. During long periods of evolution, different evolutionary rates at specific amino-acid sites can spur *Dof* family genes to evolve new function after divergence.

Thus, functional divergence analysis may reflect the existence of long-term selective pressures.

Comparative evolutionary relationships of Dof proteins of pigeonpea, soybean, rice and *Arabidopsis*

In order to gain an insight into the evolutionary relationships among the Dof proteins from pigeonpea, soybean, rice and *Arabidopsis*, a comprehensive phylogenetic tree was constructed (Fig. 7). A total of 38 identified CcDof proteins were subjected to multiple sequence alignment along with 78 soybean, 36 *Arabidopsis* and 30 rice Dof proteins, and an unrooted phylogenetic tree was constructed using NJ method, based on the alignment of a total of 182 Dof amino-acid sequences (Fig. 7). The phylogenetic tree revealed 7 major clusters designated as A, B, C, D, E, F and G. These clusters were further classified into 12 sub-clusters namely A1, A2, B1, B2, C1, C2, D, E1, E2, F1, F2 and G. The G cluster comprises of a maximum of 50 Dof members followed by A cluster with 35 members while cluster E has a minimum of 14 members. The C1 and E1 sub-clusters were species-specific cluster of rice while F1sub-cluster has soybean and pigeonpea Dof members. The close relatedness of Dof proteins of soybean and pigeonpea is based on strong genome similarity as evident from genes synteny reported recently [93]. The subclusters A2, B2, D, F1 and F2 were found to be unique as represent dicot members namely soybean, pigeonpea and *Arabidopsis*.

Based on the phylogenetic tree, several putative orthologs were identified like CcDof24/GmDof19.5 and GmDof03.2, CcDof5/GmDof07.2 and GmDof16.1, CcDof8/GmDof10.2

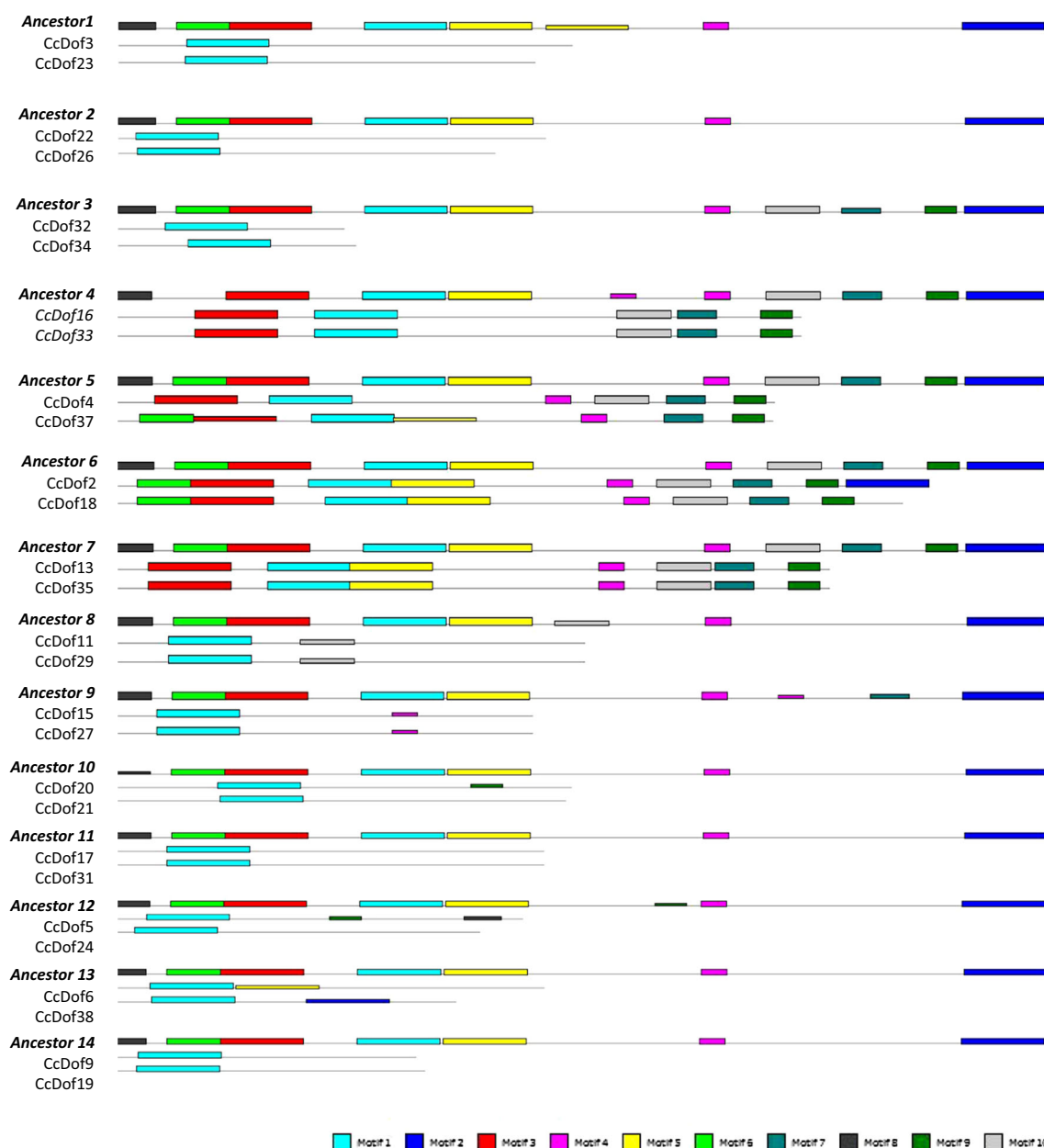


Fig. 6 Distribution of 10 motifs in gene duplicates and their ancestors

and GmDof20.2, CcDof12/GmDof04.4, CcDof6/GmDof11.3 and GmDof12.2, CcDof22/GmDof10.1 and GmDof02.4, CcDof26/GmDof19.3 and GmDof19.4, CcDof38/GmDof16.2 and GmDof02.1, CcDof1/GmDof18.4 and GmDof02.2, CcDof36/GmDof08.3 and GmDof18.1, CcDof9/GmDof08.1 and GmDof07.1, CcDof19/GmDof13.11 and GmDof15.1, CcDof3/GmDof13.4 and GmDof07.3, CcDof23/GmDof13.8 and GmDof15.4, CcDof30/GmDof13.6 and GmDof15.6, CcDof7/GmDof17.4, CcDof32/GmDof13.5 and GmDof15.7, CcDof14/GmDof19.2 and GmDof03.1, CcDof10/GmDof8.2 and GmDof15.8, CcDof28/GmDof07.6 and GmDof20.1,

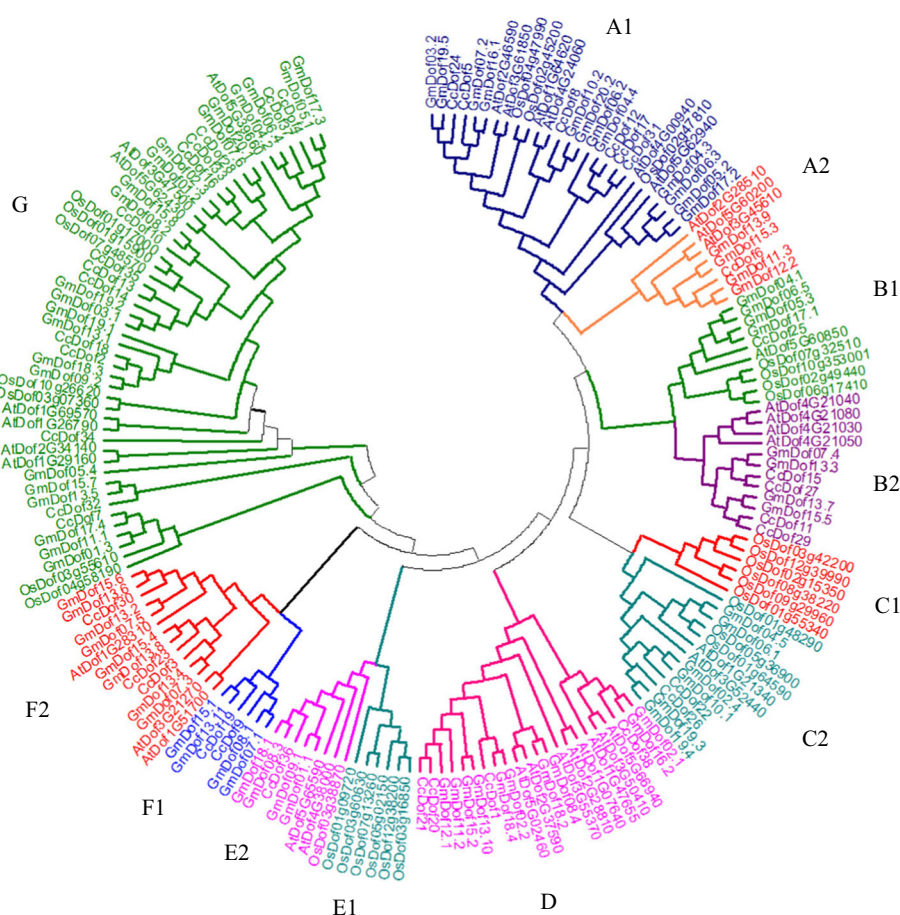
CcDof37/GmDof06.4 and GmDof04.2, CcDof4/GmDof05.1 and GmDof17.3.

The CcDof24 and CcDof5 Dof proteins representing A1 sub-cluster as revealed by phylogenetic tree might be associated with seed germination based on its maximum similarity with AtDof2G46590 and AtDof3G61850 coding for DAG2 and DAG1 protein associated with seed germination [34–37]. Similarly CcDof6 of A2 and CcDof25 of B1 sub-cluster showed maximum similarity with GmDof13.9 and GmDof17.1 of soybean respectively, known to be associated with enhancement of lipid content

Table 4 Multilevel consensus sequences for the MEME defined 10 different motifs observed among 15 pairs of paralogous Dof proteins of pigeonpea and their ancestors. The mean amino-acids are shown without square brackets

Motifs	Consensus sequence
Motif 1	L[KPN]CPRC[ND]S[TM][ND]TKFCY[YF]NNY[NS][LV][SN]QPR[HY]FCK[TN] C[RQ]RYWT[AK]GG[TSA][LM]RNVPGV[GA]G[RSC]RKNK[RKN]
Motif 2	[QE][AS]A[NG][IP][GE][SG][SN][VT][TN][SC][LK][IA]LKPHWFTSLVLKCYQLK [LM]EKGNVCLM]AIGLHLVVREKVPLKNY
Motif 3	EERTKLPSGLEELSKDEVEVP[LP]T[ED]NSE[QE]QADSSSESDSTQNPKNKSGTCEESITL
Motif 4	[AP]VP[TA]VCPPGFPL[SP]FYPA
Motif 5	S[AT]S[HS][YS][RS][SHQ][IS][ST][SV][PAS][AE][SA]A[LI]Q[TN]ARL[ED][AT][TP] [DN][SG][AS][SL][FH][LQ]LVSTLE[LS]AAI[YF][ER][SP][KL][DTK]P[SN][DG] [ST][AV]LS[SF][GS]PD
Motif 6	Q[SP]SR[PT][VD]A[SP][LA]EAPAESPSDSLDDSEQS[EH]VSS[AT]NSSR
Motif 7	[VL][LW][VI]PKTLRIDDP[PS][GNST]EA[AS]K[SC][SP][IL]W[ST]T[LS][GL][GI]
Motif 8	[ML]VFPSLP[VL][YL]LLSGNGAKM[GS][LR][SE][GSP][DQM][ED][ITV]
Motif 9	[HK][GN][HM][VS][VL][EGQ][AV]S[PQ][VI]L[QER]ANPAA[LI]SR[SAC]
Motif 10	[SC][LH]SP[TP]SS[PT][SP][NAS]S[PCT][NC]SG[PN][NG]SPTLGKH[ST]R[DE][GTD] [DNV][IF][LT][KDF][EQP][ENP][KEG][ST]

Fig. 7 Phylogenetic tree constructed using MEGA5.2.2 by NJ method based on a total of 182 Dof proteins comprising of 38pigeonpea, 78soybean, 30 rice and 36 *Arabidopsis*Dof proteins. The phylogenetic tree reveals 7 main clusters (A, B, C, D, E, F and G) divided into 12 subclusters (A1, A2, B1, B2, C1, C2, D, E1, E2, F1, F2 and G)



in the seed [94]. In cluster D, the CcDof38 showed similarity with a Dof protein of *Arabidopsis* i.e. At3G50410 associated with defense response [6]. The CcDof36 protein was found to be closer to AtDof4G38000 of *Arabidopsis* associated with induction of floral organ abscission

deficiency [41]. In cluster G, two of the Dof proteins namely CcDof4 and CcDof37 revealed maximum similarity with *Arabidopsis* Dof protein i.e. AtDof5G39660, which is a CDF2 involved in photoperiodic control of flowering [85]. The CcDof35 and CcDof13 showed

similarity with rice Dof protein i.e. OsDof01g15900 known to be associated with photoperiodic control [44]. The overall phylogenetic analysis of pigeonpea Dof proteins clearly indicates their maximum similarities with soybean Dof proteins which reflect the similarity of the genomes with extensive gene synteny as reported recently [93]. The members of the Dof proteins of pigeonpea also showed closeness with Arabidopsis Dof proteins owing to the dicot nature as compared to Dof proteins of rice representing monocot. Further, the existence of some orthologous *Dof* genes between monocot and dicot as revealed by phylogenetic tree indicates the possibility of existence of few *Dof* genes prior to the divergence of monocotyledons and dicotyledons while the existence of several paralogous genes might be due to the expansion of part of the gene family after the monocot and dicot divergence.

Conclusion

The outcome of plant genome sequencing projects has led to the deciphering of genome sequences of several plants, which needs to be annotated, prior to developing strategies for crop improvement using the genomic information. Transcription factor represents one important group targeted for crop improvement and substantial efforts have been made to reveal the whole set of transcription factors in a particular crop based on the whole genome sequence data. In the present study, efforts has been made to characterize the Dof TFs in recently sequenced pigeonpea genome using high-throughput genome-wide bioinformatics tools revealing the putative functional diversity. The comprehensive genome-wide analysis has led to an identification of 38 Dof TF proteins in pigeonpea. The multiple sequence alignment of these CcDof proteins revealed a highly conserved four cysteine residues associated with a typical Dof domain, a unique feature of Dof family of transcription factor. Further, extensive in silico characterization of these 38 Dof proteins provides an insight to the diversity of gene structure in terms of numbers and positions of exon/introns revealing functional diversity. Further the study indicates the possibility of segmental duplications of the pigeonpea genome contributing significantly to the expansion of the *CcDof* genes. The comparative phylogenetic analysis of Dof proteins of pigeonpea with Arabidopsis, rice and soybean Dof TFs has led to the identification of several orthologs and paralogs and also give an insight to the putative functions based on the close similarity with *Dof* genes with known functions in other crops.

Acknowledgments The financial support by Department of Science and Technology, Government of India, New Delhi in the form of

Women Scientist-A fellowship (SR/WOS-A/LS-110/2012(G)) to N. Malviya is thankfully acknowledged. S. Gupta acknowledges the CSIR, New Delhi for Senior Research Fellowship. The author wishes to acknowledge the Head, Department of Biotechnology, D.D.U. Gorakhpur University, Gorakhpur, INDIA for infrastructural support.

References

1. Noguero M, Atif RM, Ochatt S, Thompson RD (2013) The role of the DNA-binding one zinc finger (DOF) transcription factor family in plants. *Plant Sci* 209:32–45
2. Yanagisawa S, Izui K (1993) Molecular-cloning of 2 DNA-binding proteins of maize that are structurally different but interact with the same sequence motif. *J Biol Chem* 268:16028–16036
3. DePaolis A, Sabatini S, DePascalis L, Costantino P, Capone I (1996) A rolB regulatory factor belongs to a new class of single zinc finger plant proteins. *Plant J* 10:215–223
4. Kisu Y, Esaka M, Suzuki M (1995) Putative zinc-binding domain of plant transcription factor, AOBP, is related to DNA-binding domains of steroid hormone receptors and GATA1. *Proc Jpn Acad Ser B* 71:288–292
5. Vicente-Carbajosa J, Moose SP, Parson RL, Schmidt RJ (1997) A maize zinc-finger protein binds the prolamins box in zein gene promoters and interacts with the basic leucine zipper transcriptional activator. *Proc Natl Acad Sci USA* 94:7685–7690
6. Zhang B, Chen W, Foley RC, Buttner M, Singh KB (1995) Interactions between distinct types of DNA binding proteins enhances binding to ocs element promoter sequences. *Plant Cell* 7:2241–2252
7. Moreno-Risueno MA, Martinez M, Vicente-Carbajosa J, Carbonero P (2007) The family of DOF transcription factors: from green unicellular algae to vascular plants. *Mol Genet Genomics* 277:379
8. Lijavetzky D, Carbonero P, Vicente-Carbajosa J (2003) Genome-wide comparative phylogenetic analysis of the rice and *Arabidopsis* Dof gene families. *BMC Evol Biol* 3:17
9. Yang X, Tuskan GA, Cheng ZM (2006) Divergence of the Dof gene families in poplar, *Arabidopsis*, and rice suggests multiple modes of gene evolution after duplication. *Plant Physiol* 142:820–830
10. Shaw LM, McIntyre CL, Gresshoff PM, Xue GP (2009) Members of the Dof transcription factor family in *Triticum aestivum* are associated with light-mediated gene regulation. *Funct Integr Genomics* 9(4):485–498
11. Kushwaha H, Gupta S, Singh VK, Rastogi S, Yadav D (2011) Genome wide identification of Dof transcription factor gene family in *Sorghum* and its comparative phylogenetic analysis with rice and *Arabidopsis*. *Mol Biol Rep* 38:5037–5053
12. Guo Y, Qiu LJ (2013) Genome-wide analysis of the Dof transcription factor gene family reveals soybean-specific duplicable and functional characteristics. *PLoS ONE* 8(9):1–14
13. Jiang Y, Zeng B, Zhao H, Zhang M, Xie S, Lai J (2012) Genome-wide transcription factor gene prediction and their expressional tissue-specificities in maize. *J Integr Plant Biol* 54:616–630
14. Hernando AS, González CV, Carbonero P, Barrero SC (2012) The family of DOF transcription factors in *Brachypodium distachyon*: phylogenetic comparison with rice and barley DOFs and expression profiling. *BMC Evol Biol* 12:202–216
15. Cai X, Zhang Y, Zhang C, Zhang T, Hu T, Ye J, Zhang J, Wang T, Li H, Ye Z (2013) Genome-wide Analysis of Plant-specific

- Dof Transcription Factor Family in Tomato. *J Integr Plant Biol* 55(6):552–566
16. Yanagisawa S, Schmidt RJ (1999) Diversity and similarity among recognition sequences of Dof transcription factors. *Plant J* 17(2):209–214
 17. Yanagisawa S (2001) The transcriptional activation domain of the plant-specific Dof1 factor functions in plant, animal, and yeast cells. *Plant Cell Physiol* 42:813–822
 18. Diaz I, Martinez M, Isabel-La Moneda I, Rubio-Somoza I, Carbonero P (2005) The DOF protein, SAD, interacts with GAMYB in plant nuclei and activates transcription of endosperm-specific genes during barley seed development. *Plant J* 42:652–662
 19. RuBiol-Somoza I, Martinez M, Abraham Z, Diaz I, Carbonero P (2006) Ternary complex formation between HvMYBS3 and other factors involved in transcriptional control in barley seeds. *Plant J* 47(2):269–281
 20. Kawakatsu T, Yamamoto MP, Touno SM, Yasuda H, Takaiwa F (2009) Compensation and interaction between RISBZ1 and RPBFB during grain filling in rice. *Plant J* 59:908–920
 21. Zou X, Neuman D, Shen QJ (2008) Interactions of Two Transcriptional Repressors and Two Transcriptional Activators in Modulating Gibberellin Signaling in Aleurone Cells. *Plant Physiol* 148(1):176–186
 22. Yanagisawa S (1995) A novel DNA-binding domain that may form a single zinc finger motif. *Nucleic Acids Res* 23:3403–3410
 23. Yanagisawa S (1996) DOF DNA binding proteins contain a novel zinc finger motif. *Trends Plant Sci* 1:213–214
 24. Yanagisawa S (1997) Dof DNA binding domains of plant transcription factors contribute to multiple protein-protein interactions. *Eur J Biochem* 250:403–410
 25. Kang HG, Singh KB (2000) Characterization of salicylic acid-responsive, Arabidopsis Dof domain proteins: overexpression of OBP3 leads to growth defects. *Plant J* 21(4):329–339
 26. Diaz I, Vicente-Carbajosa J, Abraham Z, Martinez M, Isabel-LaMoneda I, Carbonero P (2002) The GAMYB protein from barley interacts with the DOF transcription factor BPBF and activates endosperm-specific genes during seed development. *Plant J* 29:401–414
 27. Mena M, Vicente-Carbajosa J, Schmidt RJ, Carbonero P (1998) An endosperm-specific DOF protein from barley, highly conserved in wheat, binds to and activates transcription from the prolamin-box of a native B-hordein promoter in barley endosperm. *Plant J* 16:53–62
 28. Mena M, Cejudo FJ, Isabel-Lamonedá I, Carbonero P (2002) A Role for the DOF Transcription Factor BPBF in the Regulation of Gibberellin-Responsive Genes in Barley Aleurone. *Plant Physiol* 130:111–119
 29. Dong G, Ni Z, Nie X, Sun Q (2007) Wheat DOF transcription factor WPBF interacts with TaQM and activates transcription of an alpha-gliadin gene during wheat seed development. *Plant Mol Biol* 63:73–84
 30. Yamamoto MP, Onodera Y, Touno SM, Takaiwa F (2006) Synergism between RPBFB DOF and RISBZ1 bZIP Activators in Regulation of Rice Seed Expression Genes. *Plant Physiol* 141:1694–1707
 31. Kushwaha H, Gupta N, Singh VK, Kumar A, Yadav D (2008) In silico analysis of PCR amplified DOF (DNA binding with one finger) transcription factor domain and cloned genes from cereals and millets. *Online J of Bioinform* 9(2):130–143
 32. Guo Y, Qin G, Gu H, Qu LJ (2009) Dof5.6/HCA2, a Dof transcription factor gene, regulates interfascicular cambium formation and vascular tissue development in Arabidopsis. *Plant Cell* 21:3518–3534
 33. Gardiner J, Sherr I, Scarpella E (2010) Expression of DOF genes identifies early stages of vascular development in Arabidopsis leaves. *Int J Dev Biol* 54:1389–1396
 34. Papi M, Sabatini S, Bouchez D, Camilleri C, Costantino P, Vittorioso P (2000) Identification and disruption of an Arabidopsis zinc finger gene controlling seed Germination. *Genes Dev* 14: 28–33
 35. Papi M, Sabatini S, Altamura MM, Hennig L, Schafer E, Costantino P, Vittorioso P (2002) Inactivation of the phloem-specific Dof zinc finger gene DAG1 affects response to light and integrity of the testa of Arabidopsis seeds. *Plant Physiol* 128: 411–417
 36. Gualberti G, Papi M, Bellucci L, Ricci I, Bouchez D, Camilleri C, Costantino P, Vittorioso P (2002) Mutations in the Dof zinc finger genes DAG2 and DAG1 influence with opposite effects the germination of Arabidopsis seeds. *Plant Cell* 14:1253–1263
 37. Gabriele S, Rizza A, Martone J, Circelli P, Costantino P, Vittorioso P (2010) The Dof protein DAG1 mediates PIL5 activity on seed germination by negatively regulating GA Biol synthetic gene AtGA3ox1. *Plant J* 61:312–323
 38. Rueda-Romero P, Barrero-Sicilia C, Gomez-Cadenas A, Carbonero P, Onate-Sanchez L (2012) *Arabidopsis thaliana* DOF6 negatively affects germination in non-after-ripened seeds and interacts with TCP14. *J Exp Bot* 63(5):1937–1949
 39. Yang CQ, Fang X, Wu XM, Mao YB, Wang LJ, Chen XY (2012) Transcriptional regulation of plant secondary metabolism. *J Integr Plant Biol* 54:703–712
 40. Yang J, Yang MF, Zhang WP, Chenc F, Shen SH (2011) A putative flowering time related Dof transcription factor gene, JcDof3, is controlled by the circadian clock in *Jatropha curcas*. *Plant Sci* 181:667–674
 41. Wei PC, Tan F, Gao XQ, Zhang XQ, Wang GQ, Xu H, Li LJ, Chen J, Wang XC (2010) Overexpression of AtDOF4.7, an Arabidopsis DOF family transcription factor, induces floral organ abscission deficiency Arabidopsis. *Plant Physiol* 153(3):1031–1045
 42. Yanagisawa S (2000) Dof1 and Dof2 transcription factors are associated with expression of multiple genes involved in carbon metabolism in maize. *Plant J* 21:281–288
 43. Skirycz A, Reichelt M, Burrow M, Birkemeyer C, Rolcik J, Kopka J, Zanol MI, Gershenzon J, Strnad M, Szopa J et al (2006) DOF transcription factor AtDOF1.1 (OBP2) is part of a regulatory network controlling glucosinolate Biosynthesis in Arabidopsis. *Plant J* 47:10–24
 44. Iwamoto M, Higo K, Takano V (2009) Circadian clock and phytochrome regulated Dof like gene, Rdd1 is associated with grain size in rice. *Plant, Cell Environ* 32:592–603
 45. Kumar R, Taware R, Gaur VS, Guru SK, Kumar A (2009) Influence of nitrogen on the expression of TaDof1 transcription factor in wheat and its relationship with photo synthetic and ammonium assimilating efficiency. *Mol Biol Rep* 36:2209–2220
 46. Sugiyama T, Ishida T, Tabei N, Shigyo M, Konishi M, Yoneyama T, Yanagisawa S (2012) Involvement of PpDof1 transcriptional repressor in the nutrient condition-dependent growth control of protonemal filaments in *Physcomitrella patens*. *J Exp Bot* 63:3185–3197
 47. Kim HS, Kim SJ, Abbasi N, Bressan RA, Yun DJ, Yoo SD (2010) The DOF transcription factor Dof5.1 influences leaf axial patterning by promoting Revoluta transcription in Arabidopsis. *Plant J* 64:524–535
 48. Negi J, Moriwaki K, Konishi M et al (2013) A Dof transcription factor, SCAP1, is essential for the development of functional stomata in Arabidopsis. *Curr Biol* 23:479–484
 49. Maesen Vander LJG (1989) *Cajanus cajan* (L.) Millsp. In: Maesen Vander LJG, Somaatmadja S (eds) Plant resources of South-East Asia No.1. Pulses. Pudoc/Prosea, Wageningen, The Netherlands., pp 39–42
 50. Varshney RK, Penmetsa RV, Dutta S et al (2010) Pigeonpea genomics initiative (PGI): an international effort to improve crop

- productivity of pigeonpea (*Cajanus cajan* L.). *Mol Breed* 26(3):393–408
51. Singh NK, Gupta DK, Jayaswal PK et al (2012) The first draft of the pigeonpea genome sequence. *J Plant Biochem Biotechnol* 21(1):98–112
 52. Guo A, He K, Liu D, Bai S, Gu S, Wei L, Luo J (2005) DATF: a database of *Arabidopsis* transcription factors. *Bioinformatics* 21(10):2568–2569
 53. Gao G, Zhong Y, Guo A, Zhu Q, Tang W, Zheng W, Gu X, Wei L, Luo J (2006) DRTF: a database of rice transcription factors. *Bioinformatics* 22(10):1286–1287
 54. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ (1990) Basic local alignment search tool. *J Mol Biol* 215(3):403–410
 55. Pruitt KD, Tatusova T, Maglott DR (2007) NCBI reference sequences (RefSeq): a curated nonredundant sequence database of genomes, transcripts and proteins. *Nucleic Acids Res* 35:61–65
 56. Stormo GD (2000) Gene-finding approaches for eukaryotes. *Genome Res* 10(4):394–397
 57. Solovyev V, Kosarev P, Seledsov I, Vorobyev D (2006) Automatic annotation of eukaryotic genes, pseudogenes and promoters. *Genome Biol* 7(1):S.10.1–S.10.12
 58. Punta M, Coghill PC, Eberhardt J et al (2012) The Pfam protein families database. *Nucleic Acids Res* 40:290–301
 59. Sigrist CJA, de Castro E, Cerutti L, Cuche BA, Hulo N, Bridge A, Bougueleret L, Xenarios I (2013) New and continuing developments at PROSITE. *Nucleic Acids Res* 41(D1):D344–D347
 60. Quevillon E, Silventoinen V, Pillai S, Harte N, Mulder N, Apweiler R, Lopez R (2005) InterProScan: protein domains identifier. *Nucleic Acids Res* 33:W116–W120
 61. Falquet L, Pagni M, Bucher P, Hulo N, Sigrist CJ, Hofmann K, Bairoch A (2002) The PROSITE database, its status in 2002. *Nucleic Acids Res* 30(1):235–238
 62. Brameier M, Krings A, Maccallum RM (2007) NucPred—predicting nuclear localization of proteins. *Bioinformatics* 23(9):1159–1160
 63. Gasteiger E, Hoogland C, Gattiker A, Duvaud S, Wilkins MR, Appel RD, Bairoch A (2005) Protein identification and Analysis Tools on the ExPASy Server. In: Walker John M (ed) *The Proteomics Protocols Handbook*. Humana Press,
 64. Thompson JD, Gibson TJ, Plewniak F, Jeanmougin F, Higgins DG (1997) The CLUSTAL_X windows interface: flexible strategies for multiple sequence alignment aided by quality analysis tools. *Nucl Acids Res* 25(24):4876–4882
 65. Tamura K, Peterson D, Peterson N et al (2011) MEGA5: Molecular Evolutionary Genetics Analysis Using Maximum Likelihood, Evolutionary Distance, and Maximum Parsimony Methods. *Mol Biol Evol* 28(10):2731–2739
 66. Saitou N, Nei M (1987) The neighbor-joining method: A new method for reconstructing phylogenetic trees. *Mol Biol Evol* 4:406–425
 67. Cannon EK, Cannon SB (2011) Chromosome visualization tool: a whole genome viewer. *Int J Plants Genomics* doi: [10.1155/2011/373875](https://doi.org/10.1155/2011/373875)
 68. Cannon SB, Shoemaker RC (2012) Evolutionary and comparative analyses of the soybean genome. *Breed Sci* 61:437–444
 69. Bailey TL, Elkan C (1994) Fitting a mixture model by expectation maximization to discover motifs in biopolymers. *Proceedings of the Second International Conference on Intelligent Systems for Molecular Biology*, AAAI Press, Menlo Park, California, 28–36
 70. Lescot M, Hais PD, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouz P, Rombauts S (2002) PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. *Nucleic Acids Res* 30(1):325–327
 71. Lavin M, Herendeen PS, Wojciechowski MF (2005) Evolutionary rates analysis of Leguminosae implicates a rapid diversification of lineages during the tertiary. *Syst Biol* 54(4):575–594
 72. Hollingsworth PM, Bateman RM, Gornall RJ (1999) *Molecular Systematics and Plant Evolution*. Publisher Taylor and Francis Limited
 73. Gu X (1999) Statistical methods for testing functional divergence after gene duplication. *Mol Biol Evol* 16(12):1664–1674
 74. Gu X (2001) Maximum-likelihood approach for gene family evolution under functional divergence. *Mol Biol Evol* 18(4):453–464
 75. Pupko T, Pe'er I, Graur D, Hasegawa M, Friedman N (2002) A branch-and-bound algorithm for the inference of ancestral amino-acid sequences when the replacement rate varies among sites: application to the evolution of five gene families. *Bioinformatics* 18(8):1116–1123
 76. Koralewski TE, Krutovsky KV (2011) Evolution of exon-intron structure and alternative splicing. *PLoS One* 6(3):e18055: 1–10
 77. Le HH, Nott A, Moore MJ (2003) How introns influence and enhance eukaryotic gene expression. *Trends Biochem Sci* 28:215–220
 78. Fedorova L, Fedorov A (2003) Introns in gene evolution. *Genetica* 118:123–131
 79. Roy SW, Gilbert W (2006) The evolution of spliceosomal introns: patterns, puzzles and progress. *Nat Rev Genet* 7:211–221
 80. Castillo-Davis CI, Mekhedov SL, Hartl DL, Koonin EV, Kondrashov FA (2002) Selection for short introns in highly expressed genes. *Nat Genet* 31:415–418
 81. Martinez M, Rubio-Somoza I, Fuentes R, Lara P, Carbonero P, Diaz I (2005) The barley cystatin gene (*Icy*) is regulated by DOF transcription factors in aleurone cells upon germination. *J Exp Bot* 56:547–556
 82. Singh K, Foley RC, Onate-Sanchez L (2002) Transcription factors in plant defense and stress responses. *Curr Opin Plant Biol* 5:430–436
 83. Ward JM, Cufr CA, Denzel MA, Neff MM (2005) The Dof transcription factor OBP3 modulates phytochrome and cryptochrome signaling in *Arabidopsis*. *Plant cell* 17:475–485
 84. Park DH, Lim PO, Kim JS, Cho DS, Hong SH, Nam HG (2003) The *Arabidopsis* COG1 gene encodes a Dof domain transcription factor and negatively regulates phytochrome signaling. *Plant J* 34:161–171
 85. Fornara F, Panigrahi KC, Gissot L, Sauerbrunn N, Ruhl M, Jarillo JA, Coupland G (2009) *Arabidopsis* DOF transcription factors act redundantly to reduce CONSTANS expression and are essential for a photoperiodic flowering response. *Dev Cell* 17:75–86
 86. Imaizumi T, Schultz TF, Harmon FG, Ho LA, Kay SA (2005) FKF1 F-box protein mediates cyclic degradation of a repressor of CONSTANS in *Arabidopsis*. *Science* 309:293–297
 87. Li D, Yang C, Li X, Gan Q, Zhao X, Zhu L (2009) Functional characterization of rice OsDof12. *Planta* 229:1159–1169
 88. Yang Z, Bielawski JP (2000) Statistical methods for detecting molecular adaptation. *Trends Ecol Evol* 15(12):496–503
 89. Hurst LD (2002) The Ka/Ks ratio: diagnosing the form of sequence evolution. *Trends Genet* 18(9):486–487
 90. Morgan CC, Loughran NB, Walsh TA, Harrison AJ, O'Connell MJ (2010) Positive selection neighboring functionally essential sites and disease—implicated regions of mammalian reproductive proteins. *BMC Evol Biol* 10:39
 91. Khan AA, Janke A, Shimokawa T, Zhang H (2011) Phylogenetic analysis of kindlins suggests subfunctionalization of an ancestral unduplicated kindlin into three paralogs in vertebrates. *Evol Bioinform Online* 7:7–19
 92. Mallika V, Sivakumar KC, Soniya EV (2011) Evolutionary Implications and Physicochemical Analyses of Selected Proteins

- of Type III Polyketide Synthase Family. *Evol Bioinform Online* 7:41–53
93. Varshney RK, Chen W, Li Y, Bharti AK, Saxena RK et al (2012) Draft genome sequence of pigeonpea (*Cajanus cajan*), an orphan legume crop of resource-poor farmers. *Nat Biotechnol* 30:83–89
94. Wang H, Zhang B, Hao Y, Huang J, Tian A, Liao Y, Zhang J, Chen S (2007) The soybean Dof-type transcription factor genes, GmDof4 and GmDof11, enhance lipid content in the seeds of transgenic Arabidopsis plants. *Plant J* 52:716–729